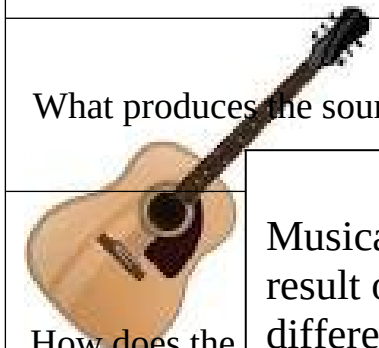
Sound Production in Musical Instruments

	String Instruments	Wind Instruments	Percussion Instruments
What produces the sound?			
How does the the vibra			
How can th chang			
Five examples of the type of instrument.			



Musical Instrument Families

Musical instruments are classified by the way they make sounds. Sounds are the result of vibrations and in each type of instrument the vibrations come from a different source.

String Instruments

Vibrating strings provide the sound in stringed instruments. The player makes the strings vibrate in one of three ways: plucking, bowing or hitting.

How the pitch is changed

The length, thickness and tension of the string all affect its pitch. Longer, thicker or looser strings will vibrate more slowly than shorter, thinner or tighter strings. Faster vibrations will cause a note that is high in pitch and slow vibrations will cause notes that are low in pitch. The different pitches on stringed instruments are obtained by having many strings of different lengths or by changing the vibrating length of the string by stopping them at different points.





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Wind Instruments

Sound produced by wind instruments comes from a vibrating column of air inside the instrument. There are two types of wind instruments, brass and woodwind. For brass instruments, the column of vibrating air is caused by the vibrating lips of the player. The vibrating column of air in woodwinds can be created in one of three ways: by blowing across an edge, like the flute, by blowing between a reed and a fixed surface, as in the clarinet or by blowing between two reeds, as in the oboe.

How the pitch is changed

Wind instrument pitch depends on the volume of air that is vibrating. A larger volume of air vibrates more slowly than a smaller volume causing a lower pitch. The player can change the pitch by opening and closing holes along the instruments length





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Percussion Instruments

In percussion instruments the source of sound is a vibrating membrane or a vibrating piece of solid material. The percussionist normally causes the materials to vibrate by hitting them but many are played by shaking or rubbing.

How the pitch is changed

Due to the complex ways the sound source vibrates, most percussion instruments do not have a definite pitch. The pitch of these instruments depends on the amount of material that is vibrating. This means that the instruments must have a different vibrating body for each note. The pitch of percussion instruments with membranes also depends on the thickness and tension in the membrane.

